Homework 4 Report — MLE and MAP

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1. Problem Setup

Given a dataset $\mathcal{D} = \{X, Y\}$ with $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$, we model:

$$y_i = \theta_1 + \theta_2 x_i + \cdots + \theta_K x_i^{K-1} + e_i$$

The feature map uses the classical Vandermonde matrix:

$$\Phi(X)_{ij} = \phi_j(x_i) = x_i^{j-1}, \quad i = 1 \dots N,\ j = 1 \dots K$$

So the model becomes $Y = \Phi(X)\theta + e$.

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2. MLE — Gaussian Noise

Assuming $e_i \sim \mathcal{N}(0, \sigma^2)$, the conditional likelihood is:

$$p_\theta(y_i|x_i) = \mathcal{N}(f_\theta(x_i), \sigma^2)$$

Maximising the log-likelihood is equivalent to minimising the least squares loss:

$$\theta_{\text{MLE}} = \arg\min_\theta \frac{1}{2\sigma^2} \|\Phi(X)\theta - Y\|_2^2$$

Closed-form solution (Normal Equations):

$$\theta_{\text{MLE}} = (\Phi^T \Phi)^{-1} \Phi^T Y$$

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3. MAP — Gaussian Prior

Assuming a Gaussian prior $\theta \sim \mathcal{N}(0, \sigma_\theta^2 I)$ and applying Bayes' theorem:

$$\theta_{\text{MAP}} = \arg\min_\theta \sum_{i=1}^{N} -\log p(y_i|x_i, \theta) - \log p(\theta)$$

This gives the Ridge Regression problem:

$$\theta_{\text{MAP}} = \arg\min_\theta \frac{1}{2\sigma^2} \|\Phi(X)\theta - Y\|^2 + \frac{\lambda}{2} \|\theta\|^2$$

where $\lambda = \sigma^2 / \sigma_\theta^2$ controls the regularisation strength.

Closed-form solution:

$$\theta_{\text{MAP}} = (\Phi^T \Phi + \lambda I)^{-1} \Phi^T Y$$

4. Solvers Implemented

Three optimisation methods were implemented and compared:

Solver	Update rule
Normal Equations (NE)	Direct closed-form matrix solve
Gradient Descent (GD)	$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ (full batch)
SGD	Same but over random mini-batches of size B

For GD on the MLE loss:

$$\nabla_\theta \mathcal{L} = \frac{1}{N} \Phi^T (\Phi\theta - Y)$$

For MAP (GD):

$$\nabla_\theta \mathcal{L} = \frac{1}{N} \Phi^T (\Phi\theta - Y) + \lambda\theta$$

5. Experiments — Gaussian Model

5.1 Training/Test Error vs K

- For $K < K_{\text{true}}$: underfitting — both train and test error are high.
- For $K = K_{\text{true}}$: best generalisation.
- For $K > K_{\text{true}}$: MLE overfits (test error rises sharply); MAP remains stable thanks to regularisation.

5.2 Regression Curves

- MLE with $K \gg K_{\text{true}}$ produces highly oscillatory fits (Runge's phenomenon).
- MAP with appropriate λ suppresses oscillations even at large K.
- Large λ leads to underfitting (over-smoothed curves).

5.3 Parameter Error vs K

For $K > K_{\text{true}}$, θ_{true} is zero-padded and:

$$\text{Err}(\theta) = \frac{||\theta - \theta_{\text{true}}||_2}{||\theta_{\text{true}}||_2}$$

- MLE parameter error grows quickly with K (ill-conditioning of $\Phi^T\Phi$).
- MAP controls this growth; smaller λ → closer to MLE behaviour, larger λ → more shrinkage toward zero.

5.4 Effect of N

- With more training data, MLE becomes more reliable even for large K (variance decreases).
- MAP advantage shrinks as N grows — consistent with the asymptotic equivalence of MLE and MAP for large datasets.

5.5 Solver Comparison

All three solvers converge to the same solution when properly tuned:

- Normal Equations: exact, fast for small K, numerically unstable for very large K.
- GD: reliable, slower, requires tuning of learning rate.
- SGD: fastest per epoch, noisier convergence, needs more epochs.

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6. Poisson Noise Model

When observations follow $y_i \sim \mathrm{Poi}(\lambda_i)$ with $\lambda_i = f_\theta(x_i)$:

$$p_\theta(y_i|x_i) = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}$$

MLE (Poisson)

$$\theta_{\text{MLE}} = \arg\min_\theta \sum_{i=1}^{N} [-y_i \log \lambda_i + \lambda_i]$$

Gradient:

$$\nabla_\theta \mathcal{L} = \frac{1}{N} \Phi^T \left(1 - \frac{Y}{\lambda}\right)$$

MAP (Poisson + Gaussian Prior)

$$\theta_{\text{MAP}} = \arg\min_\theta \sum_{i=1}^{N} [-y_i \log \lambda_i + \lambda_i] + \frac{\lambda}{2} \|\theta\|^2$$

Both solved via Gradient Descent (no closed form due to the Poisson NLL).

Observations

- Poisson MLE behaves similarly to Gaussian MLE in terms of overfitting at large K.
- MAP regularisation is equally effective at controlling variance.
- The Poisson model is more appropriate when y_i are count data (non-negative integers).

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7. Conclusions

Setting	**MLE**	**MAP**
$K = K_{\text{true}}$	Good fit	Good fit (λ matters less)
$K > K_{\text{true}}$, small N	Overfits	Stable with right λ
$K > K_{\text{true}}$, large N	Recovers	≈ MLE
Poisson noise	Works via GD	Works via GD + prior

Key takeaway: **MAP = MLE + regularisation**. When K is unknown and N is limited, MAP with cross-validated λ is the safer choice.